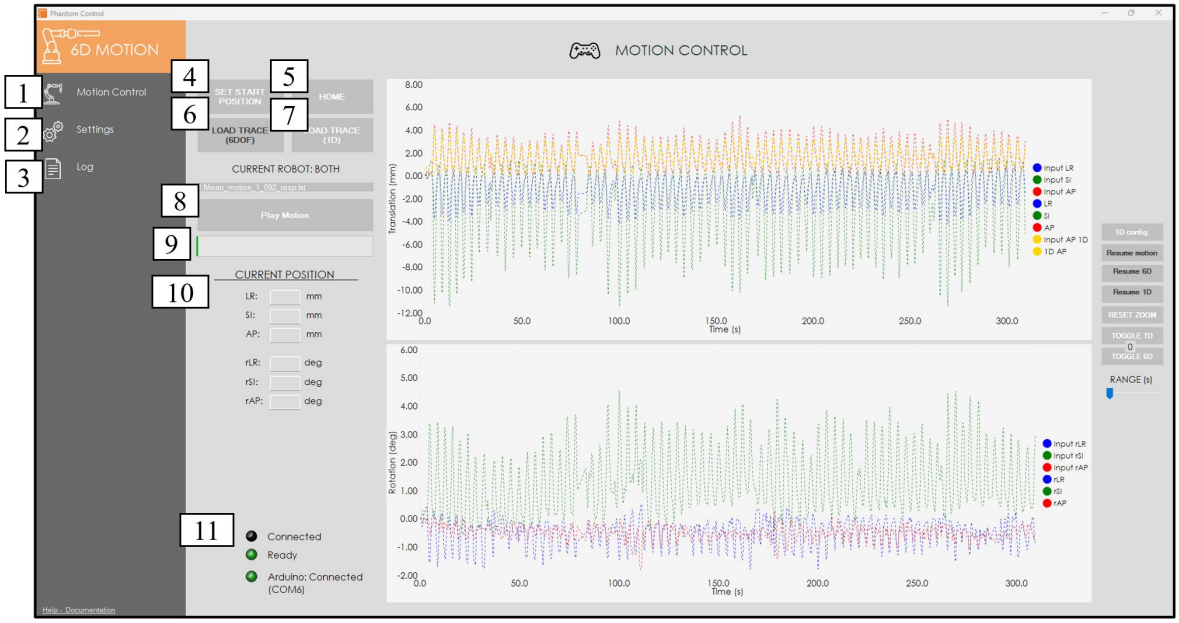
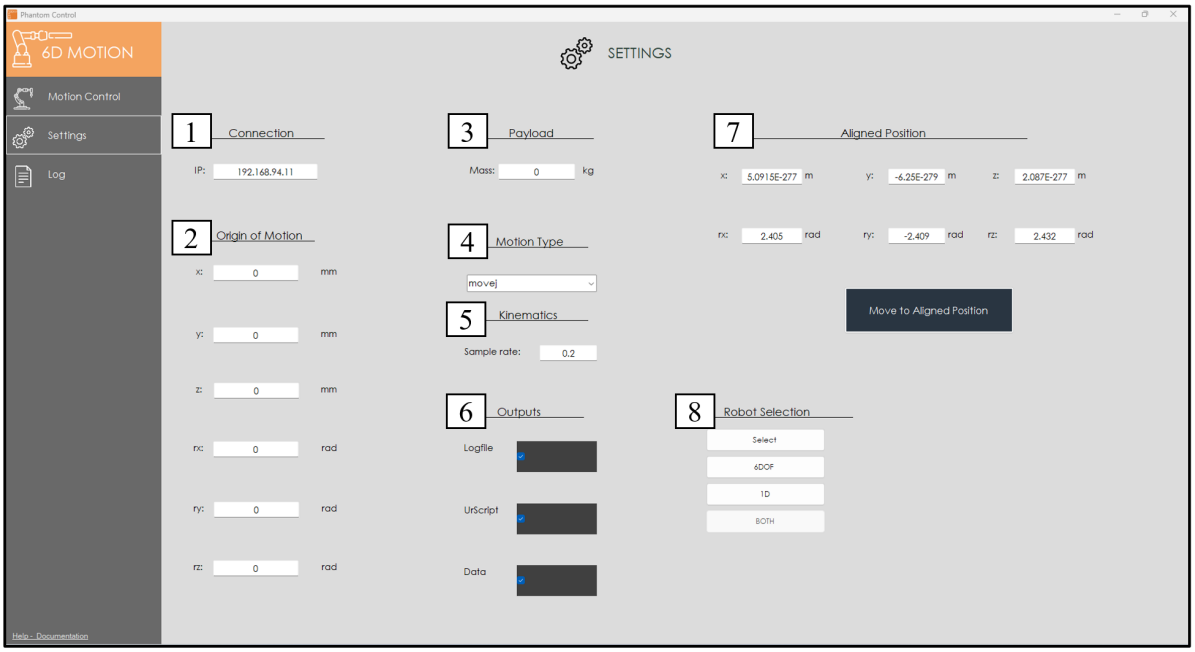
IntERAct QA Device
Software GUI Guide



1. **Motion control tab**
   Opens controls to load traces and run motion (current tab in image above)
2. **Settings tab**
   Opens settings for IntERAct motion platform
3. **Log tab**
   Opens log box to view status of IntERAct motion platform
4. **Set start position**
   Once the 6DoF robotic arm is setup and aligned, click the *Set Start Position* button. This will first establish a connection with the 6DoF robot and get the current (aligned) position of the 6DoF platform.
5. **Home**
   The *Home* button will send the 6DoF motion platform back to the original start position that was set by the *Set Start Position* button. If the 6DoF robot is repositioned and the *Set Start Position* button is clicked again the new *Home* button will be updated to that position.
6. **Load motion trace (6DoF)**
   Once the *Set Start Position* button is clicked and a connection to the 6DoF robot is established, the *Load Motion Trace (6DoF)* button will become active. This will allow a user to load a .txt file in the following format [Time (s) x (mm) y (mm) z (mm) rx (deg) ry (deg) rz (deg)] (space delimited file). The loaded trace will then be displayed on the graph on the right.
7. **Load motion trace (1DoF)**
   Once the *Set Start Position* button is clicked and a connection to the 1DoF motion platform is established, the *Load Motion Trace (1DoF)* button will become active. This will allow a user to load a .txt file in the format [Time (s) z (mm)] (space delimited file). The initial z position needs to be 0 (corresponding to the fully retracted position of the 1DoF platform). The loaded trace will then be displayed on the graph on the right.
8. **Play motion**
   Once the loaded trace(s) is complete and the transformed motion is (automatically) checked for irregular motion (singularity check, large motion check), the *Play Motion* button will become active. Clicking this will send the data to the robot and run the motion. The *Play Motion* button will then change to a *Stop Motion* icon – clicking this will stop the IntERAct motion platform from moving.
9. **Status bar**
   Provides an estimated status of progress of the motion that is running.
10. **Current position value box**

Displays the current position of the 6DoF robot when a motion trace is running.

11. **Display LEDs**
    Provide information if the 6DoF and 1DoF motion platforms are connected and ready for motion. If there is an issue with the motion e.g. a singularity or very large motion, the Ready LED will be red, and the *Play Motion* button will become inactive. If all LEDs are green motion can be run.



1. **Connection**
   Input the IP address of the 6DoF robotic controller for TCP/IP connection.
2. **Origin of motion**
   Sets the origin of motion for both translation and rotation for the input 6DoF robot traces relative to the centre of the end-effector. This is known as the Tool Centre Point (TCP).
3. **Payload**
   Input the weight of the phantom attached to the 6DoF robot in kg including the tool attachment to the end-effector. This is important for the safety of the 6DoF robot and accurate motion.
4. **Motion type**
   Two motion types can be run on the 6DoF robot: movej (default) and movel. Movej makes the robotic arm move from point a to point b in the most efficient way when moving the joints in the arm. Movel makes the robotic arm move from point a to point b linearly, i.e. the joints will move in a way that will force the robot to move in a straight path.
5. **Kinematics**
   The input file sample rate defines the velocity and acceleration of the 6DoF robot. The input sample rate should be the same as the input data sampling frequency (5Hz/0.2 s intervals recommended). This will allow the robot to move from point a to point b in at the set input sampling time.
6. **Outputs**
   Output files can be switched on and off. Logfile outputs a .txt of the logging information during a motion run. UrScript outputs what is sent to the 6DoF robotic controller for executed motion, this is in the robot API language. Data outputs the motion recorded from the 6DoF robot as a .txt file in the format [Time (s) x (mm) y (mm) z (mm) rx (deg) ry (deg) rz (deg)] (space delimited).
7. **Aligned position**
   Once the 6DoF robot (specifically, any attached phantom) is aligned to the treatment isocentre, the user may record the 6DoF position from the tablet of the robot in the Base coordinate system.
8. **Robot selection**
   Setting to select the 6DoF motion platform, 1DoF motion platform, or the combined system ('both')